FLOODGAUGE · DESIGN DECISION RECORD

FloodGauge TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	FLG-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/decisions/0001-trl2-review-decisions.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 decided by Amish on 2026-10-02 (FLG-DEC-001, item 6)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 work, open for Amish's review, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted. On 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos", so items D1 to D9 are "Decided by Amish, 2026-09-25: go with recommendation" (recorded in FLG-DDR-002). Item O1 had no recommendation at this record; it was decided by Amish on 2026-10-02 (FLG-DEC-001, item 6): "i approve your recommendations for all 555 open decisions."

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis FLG-PRC-001 v0.2 listed seven key design choices, all marked proposed. On 2026-09-25 Amish asked for this batch of Design Molecule repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carried a recommendation is therefore adopted as recommended for TRL 3 work, open for his review. The item without a recommendation stays open. Nothing here is recorded as decided or approved by Amish.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (session 2026-09-25, /populate) and in FLG-PRC-001 v0.2, Key design choices.

Decision

Table 1. Items decided (first adopted for TRL 3 work, then decided by Amish on 2026-09-25).

#	Item	Adopted recommendation	Status
D1	Budget (review item 1)	Option (a): cost the FieldNode core in the FieldNode repo and hold FloodGauge-specific parts to $150. R16 is restated as "FloodGauge-specific parts $150 or less; FieldNode costed separately". `budget_usd` stays $150; no new budget figure was recommended	Decided by Amish, 2026-09-25: go with recommendation
D2	Heads (review item 2, precis choice 1)	Two heads on one node: street and drain; street head only is the fallback where no cable route to the basin exists	Decided by Amish, 2026-09-25: go with recommendation
D3	Street head sensing (review item 3, precis choice 2)	60 GHz pulsed coherent radar; an ultrasonic street head is documented as the lower-cost variant	Decided by Amish, 2026-09-25: go with recommendation
D4	Drain head (review item 4, precis choice 3)	Waterproof ultrasonic ranger in a slotted 75 mm stilling tube, with a two-electrode wet probe as a second signal	Decided by Amish, 2026-09-25: go with recommendation
D5	Head position (review item 5, precis choice 4)	Arm over the gutter with the head at 2.95 m or higher, subject to the road authority's clearance rules	Decided by Amish, 2026-09-25: go with recommendation
D6	Alert bands (review item 6, precis choice 5)	150 mm and 300 mm of water above the road, from NWS guidance, plus a drain-full alert from the wet probe; defaults to be set locally with the partner city	Decided by Amish, 2026-09-25: go with recommendation
D7	Network (review item 7, precis choice 6)	Private or city LoRaWAN gateway (TwinKit or the city network) for event reporting; The Things Network for pilots only	Decided by Amish, 2026-09-25: go with recommendation
D8	Privacy (precis choice 7)	Levels only; no camera, no microphone, no imaging	Decided by Amish, 2026-09-25: go with recommendation
D9	Pitch and problem (review item 9)	No change was recommended; the pitch and problem lines in `project.yaml` and `README.md` are unchanged	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remain open.

#	Item	Status
O1	Pilot partner and city for co-design and a first site (review item 8). No preference stated and no recommendation made.	Decided by Amish, 2026-10-02 (FLG-DEC-001, item 6): a city or county flood agency with street flooding at grated curb inlets and an existing gauge or alert programme; first candidate type to approach, a county flood control district such as the Harris County Flood Control District; first alert band 100 mm unless the partner's practice says otherwise. Nothing is agreed.

Consequences

- `project.yaml` : only the TRL fields and the evidence list change. `budget_usd` stays $150, and the pitch and problem are unchanged (D9).

- FLG-PRB-001, FLG-PRC-001 and FLG-REQ-001 are revised to v0.3. The design choices in the precis are no longer described as "proposed"; they are adopted for TRL 3 work pending Amish's review.
- Requirement R16 is restated to match D1: it now covers FloodGauge-specific parts only, with the FieldNode core costed in its own repo ($126.00, FND-CAL-001). No other target changes in this record.
- `bom/bom.csv` keeps the FieldNode core as line 1 for the complete-gauge total, marked as costed in the FieldNode repo and excluded from the R16 total.
- The TRL 3 calculations (FLG-CAL-001) found problems that need Amish's decision: R3 is not met above 330 mm below the road, R12 is not met because of the conduit, the 2.95 m head height is inside a tall vehicle envelope, and the arm's twist margin on two band clamps is 1.45. The proposals for these are in `docs/REVIEW.md` and are not decided by this record.
- The later problems listed above were also decided on 2026-09-25; see FLG-DDR-002.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.